

Nuclear Physics Angular Momentum Coupling Guide

This document provides the theoretical background and practical usage guide for the `angular_momentum_coupling.py` utility. This tool calculates allowed final nuclear states based on quantum mechanical selection rules.

1. Theoretical Basis

The code implements the **Channel Spin Coupling Scheme**, which is a standard method for determining angular momentum conservation in nuclear reactions.

Conservation Laws

In any nuclear reaction $Target + Particle \rightarrow Final_State$, the following quantities are conserved:

1. **Total Angular Momentum** ($\vec{J}$):

$$\vec{J}_{final} = \vec{J}_{target} + \vec{s}_{particle} + \vec{\ell}$$

2. **Parity** (π):

$$\pi_{final} = \pi_{target} \times \pi_{particle} \times (-1)^\ell$$

Where:

- $\vec{J}_{target}$: Spin of the target nucleus.
- $\vec{s}_{particle}$: Intrinsic spin of the interacting particle (projectile or transferred cluster).
- $\vec{\ell}$: Relative orbital angular momentum between the target and the particle.

Coupling Scheme Used in Code

The code performs the vector addition in two steps (Channel Spin representation):

1. **Calculate Channel Spin** ($\vec{S}$): Couples the target spin and particle spin.

$$\vec{S} = \vec{J}_{target} + \vec{s}_{particle}$$

Possible values: $|J_{target} - s_{particle}| \leq S \leq J_{target} + s_{particle}$

2. **Calculate Final Spin** ($\vec{J}_{final}$): Couples the channel spin with the orbital angular momentum.

$$\vec{J}_{final} = \vec{S} + \vec{\ell}$$

Possible values: $|S - \ell| \leq J_{final} \leq S + \ell$

2. Application to Different Reaction Types

To use the code correctly, you must identify the correct "Particle" input based on the reaction mechanism.

A. Resonance and Capture Reactions

Examples: (p, γ) , (n, γ) , (p, p')

In these reactions, a projectile fuses with the target to form a compound nucleus (resonance).

- **Target Input:** Spin/Parity of the target nucleus.
- **Particle Input:** Spin/Parity of the **projectile**.

Reaction	Projectile	Particle Input
$^{59}\text{Cu}(p, \gamma)^{60}\text{Zn}$	Proton	1/2+
$^{12}\text{C}(n, \gamma)^{13}\text{C}$	Neutron	1/2+
$^{16}\text{O}(\alpha, \gamma)^{20}\text{Ne}$	Alpha	0+

B. Single-Nucleon Transfer Reactions

Examples: (d, p) , (p, d) , $(^3\text{He}, d)$, (α, t)

In these reactions, a single nucleon is transferred between the projectile and the target. You must input the properties of the **transferred nucleon**, not the beam.

- **Target Input:** Spin/Parity of the target nucleus.
- **Particle Input:** Spin/Parity of the **transferred nucleon** (n or p).

Reaction Type	Example	Transferred Object	Particle Input
Pickup	$^{13}\text{C}(p, d)^{12}\text{C}$	Neutron	1/2+
Stripping	$^{40}\text{Ca}(^3\text{He}, d)^{41}\text{Sc}$	Proton	1/2+

Note on j-transfer: In transfer physics, we often discuss the transferred total angular momentum j (where $\vec{j} = \vec{\ell} + \vec{s}$). The code lists results by Channel Spin S . The set of allowed final J^π values is identical regardless of the coupling order.

C. Two-Nucleon Transfer Reactions

Examples: (p, t) , (t, p) , $(^3\text{He}, n)$, (α, d)

In these reactions, a pair of nucleons is transferred. The total spin of this pair depends on the coupling of the two nucleons.

1. Identical Nucleon Transfer (2n or 2p)

Reactions: (p, t) , (t, p) (2n transfer); $(^3\text{He}, n)$ (2p transfer)

For the transfer of two identical nucleons ($2n$ or $2p$) in the same shell model orbit, the **Pauli Exclusion Principle** dictates that their total wavefunction must be antisymmetric.

- If they are in a relative s-state (spatially symmetric, $L = 0$), their **spin wavefunction must be antisymmetric**.
- An antisymmetric spin state for two fermions corresponds to a **Singlet State** ($S = 0$).
- **ENSDF Policy:** For (p, t) , (t, p) , and $({}^3\text{He}, n)$ reactions, it is standard to assume the transferred pair is in an **anti-parallel spin state** ($S = 0$).
- **Particle Input:** $0+$

2. Neutron-Proton Transfer ($1n1p$)

Reactions: (α, d) , (d, α)

For the transfer of a neutron-proton pair (a deuteron), the Pauli principle does not restrict the spin to $S = 0$. The deuteron itself has spin $S = 1$.

- **Spin Configurations:** Both singlet ($S = 0$) and triplet ($S = 1$) transfers are possible.
- **Particle Input:**
 - If assuming singlet transfer: $0+$
 - If assuming triplet transfer (e.g., deuteron-like): $1+$
 - *Note: Selection rules often require analyzing both possibilities.*

Transfer Type	Transferred Pair	Spin Assumption	Particle Input
$2n / 2p$	Identical	Anti-parallel ($S = 0$)	$0+$
$1n1p$	Non-identical	Parallel ($S = 1$) or Anti-parallel ($S = 0$)	$1+$ or $0+$

D. Cluster Transfer Reactions

Examples: $({}^6\text{Li}, d)$, $({}^7\text{Li}, t)$

An alpha particle (or other cluster) is transferred.

- **Target Input:** Spin/Parity of the target nucleus.
- **Particle Input:** Spin/Parity of the **transferred cluster**.

Reaction	Transferred Cluster	Particle Input
$({}^6\text{Li}, d)$	Alpha (${}^4\text{He}$)	$0+$

E. Alpha Inelastic Scattering

Examples: (α, α')

In inelastic scattering, the target nucleus is excited to a higher energy state.

- (α, α') : The alpha particle has spin 0. Since the projectile spin cannot flip, the angular momentum transfer is purely orbital ($\vec{\ell}$). This selectively excites **Natural Parity** states ($\pi = (-1)^\ell$).
 - **Particle Input:** 0+
 - *Note: This treats the angular momentum transfer as a spin-0 boson transfer.*
-

3. Interpreting the Output

When you run the code:

```
python angular_momentum_coupling.py 3/2- 1/2+
```

You receive output like:

L	Wave	Parity	s	Final J π
0	s	-	1	1-
0	s	-	2	2-

- **L (Wave):** The orbital angular momentum of the incoming/transferred particle.
 - $L = 0$ (s-wave), $L = 1$ (p-wave), etc.
- **Parity:** The parity of the final state, determined by $\pi_{final} = \pi_{target} \times \pi_{particle} \times (-1)^\ell$.
- **s (Channel Spin):** The intermediate coupling of Target + Particle.
 - For a $3/2^-$ target and $1/2^+$ particle, spins can align ($3/2 + 1/2 = 2$) or anti-align ($3/2 - 1/2 = 1$).
- **Final J π :** The allowed total angular momentum and parity of the final state.

Example Analysis: $^{59}\text{Cu}(p, \gamma)^{60}\text{Zn}$

- **Target:** ^{59}Cu ($3/2^-$)
- **Particle:** Proton ($1/2^+$)
- **Result:**
 - If capture occurs via **s-wave** ($L = 0$):
 - Channel spins $S = 1, 2$.
 - Final states $J^\pi = 1^-, 2^-$.
 - If capture occurs via **p-wave** ($L = 1$):
 - Parity flips ($-$ to $+$).
 - Final states include $0^+, 1^+, 2^+, 3^+$.